

Monthly Intelligence Report

Edition 003 — The Andalucía-Murcia Seismic Corridor

Where the energy transition is outrunning the grid. edition · June 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

3,529
939 EHV · 2,590 HV

GRID LINES MAPPED

~22,765
~87,000 km coverage

MC SIMULATIONS

35.3 M
10,000 per substation

PUBLIC DATA SOURCES

30+
95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 65,300+ source data points per run, executes 35.3 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 794 million arithmetic operations — producing 209,722 output data points with full uncertainty quantification across 3,529 substations and ~22,765 grid lines spanning ~105,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Spain's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

SET Novallas

Zaragoza, Aragón · 66 kV

0.750

R_FINAL

Critical

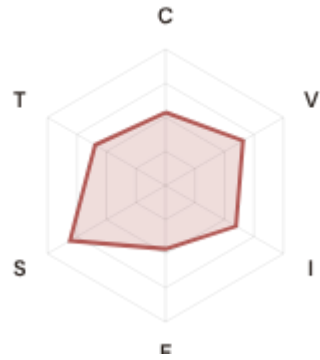
CLASSIFICATION

0.241

CI WIDTH

73th

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.536 / 0.30 (179%)

I Infrastructure: 0.598 / 0.25 (239%)

S Saturation: 0.809 / 0.20 (405%) **▲**

V Voltage: 0.657 / 0.10 (657%) **▲**

E Economic: 0.465 / 0.10 (465%)

T Transition: 0.596 / 0.05 (1192%)

MODIFIERS

R3 Consequence: 0.970 ↓ dampens

R4 Graph Criticality: 1.020 ↑ amplifies

R6a Restoration: 0.992 → neutral

R6b Seismic: 1.249 ↑ amplifies

R7 Digital Readiness: 0.997 → neutral

This month's spotlight — automatically selected for June 2026 — is **SET Novallas** in Zaragoza, Aragón. With an R_final of 0.750 (Critical band, 73th fleet percentile), its risk profile is dominated by the T (Transition) component at 1192% of its weight ceiling, followed by V (Voltage) at 657%. Modifiers collectively shift R_base by 22.2%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Spain's Plan de Digitalización / PNIEC 2030 planning requires.

CYBER HIGH EXPOSURE

633

17.9% of fleet

BLIND SPOTS

1630

High/Critical + R7 < median

AVG R7 MODIFIER

1.0079

Range [0.99, 1.05]

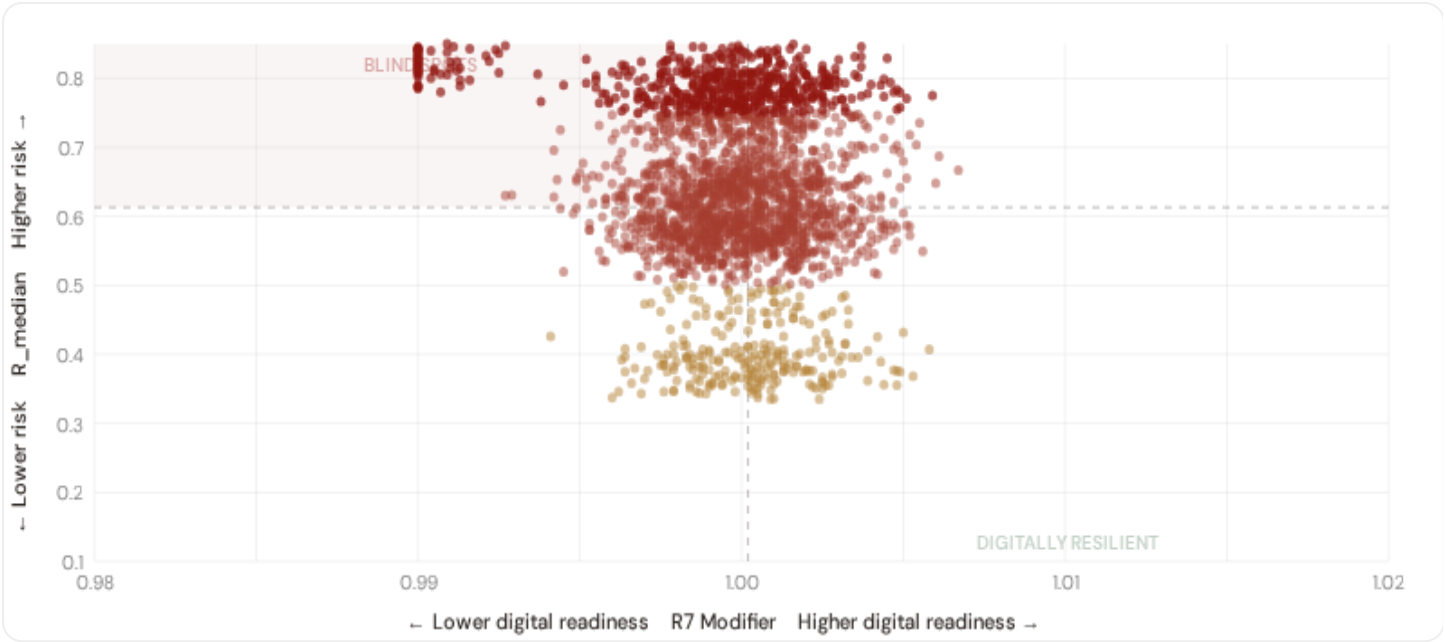
HIGH-CONSEQUENCE SUBS

1,594

45.2% · R3 ≥ 1.13

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

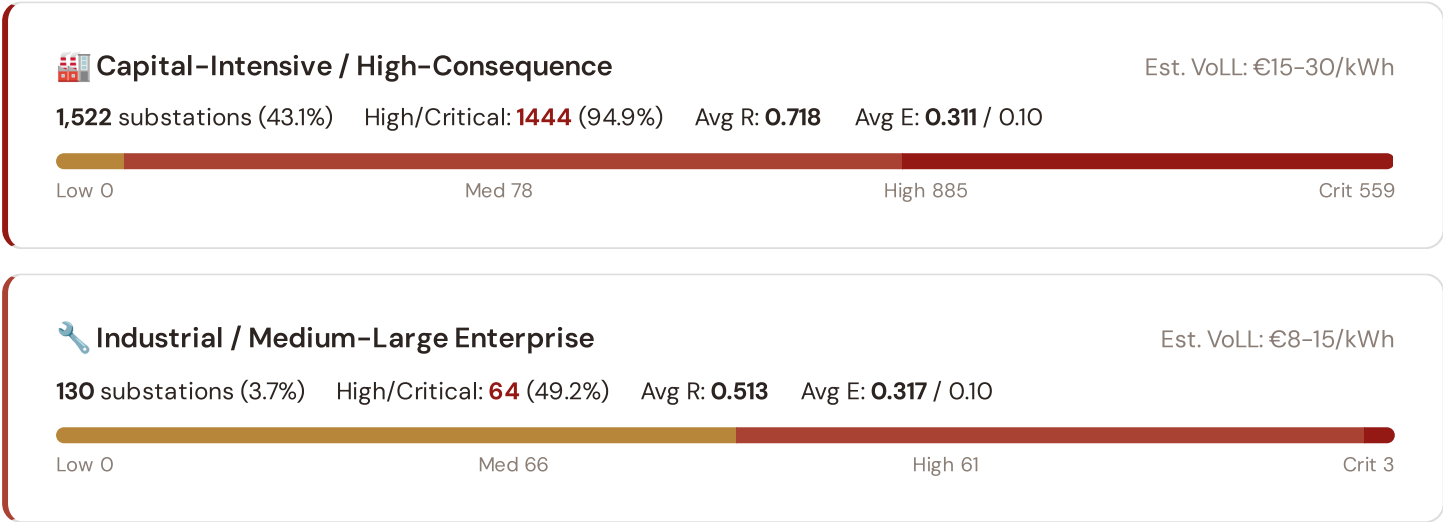


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **1630 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0002$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Andalucía (345)**, **Castilla-La Mancha (334)**, **Castilla y León (269)**. Across the full fleet, 633 substations (17.9%) carry a HIGH cyber-exposure classification, while 2782 (78.8%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Commercial / Mixed

Est. VoLL: €3–8/kWh

122 substations (3.5%) High/Critical: **67** (54.9%) Avg R: **0.533** Avg E: **0.347** / 0.10



Light-Rural / Agricultural

Est. VoLL: €1–3/kWh

1,755 substations (49.7%) High/Critical: **969** (55.2%) Avg R: **0.537** Avg E: **0.364** / 0.10



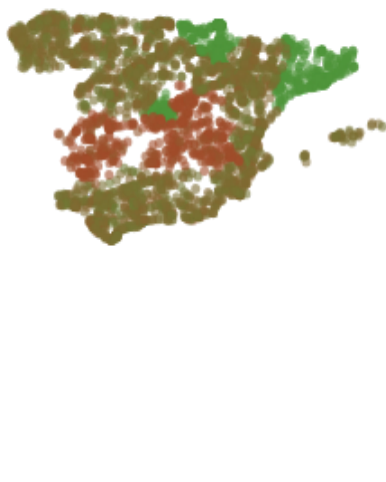
Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **1444 substations** combine capital-intensive economic fabric ($R3 \geq 1.14$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Andalucía (606)**, **Castilla-La Mancha (334)**, **Extremadura (173)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 10.6%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Provincia Digital Readiness Ranking

Provincias ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify provincias combining digital exposure with high grid stress.

R7 Digital Readiness
Low High



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Toledo Δ high R		0.9907
2	Guadalajara Δ high R		0.9908
3	Badajoz Δ high R		0.9908
4	Cuenca Δ high R		0.9908
5	Ciudad Real Δ high R		0.9909
6	Albacete Δ high R		0.9909
7	Cáceres Δ high R		0.9912
8	Almería Δ high R		0.9993
9	Cádiz Δ high R		0.9995
10	Jaén		0.9995
11	Castellón		0.9998
12	Santa Cruz de Tenerife Δ high R		0.9998

13	Las Palmas ▲ high R		0.9998
14	León ▲ high R		0.9998
15	Alicante		0.9999

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Toledo** has the lowest average R7 (0.9907), while **Álava** leads at 1.0495 — a spread of 0.0588 across the R7 range. **21 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0079, median = 1.0002; 633 substations classified HIGH cyber-exposure; 1630 blind-spot substations (High/Critical risk + below-median R7); 756 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

30

95 variables

WEEKLY / MONTHLY

4

High-frequency feeds

QUARTERLY / ANNUAL

18

Regular refresh cycle

STATIC / DERIVED

8

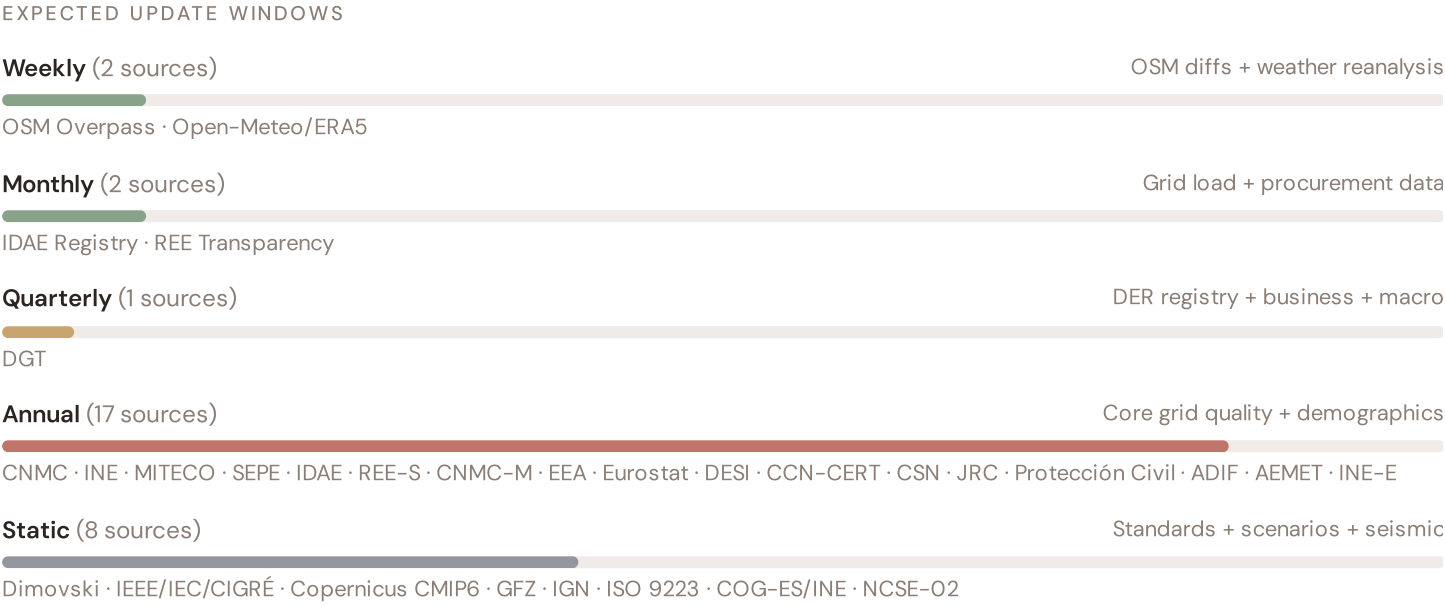
Standards + scenarios

Data Source Registry — June 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
SEPE	SEPE (Servicio Público de Empleo Estatal)	MONTHLY	Provincia	1 var
IDAE	IDAE (Instituto para la Diversificación y Ahorro de la Energía)	QUARTERLY	Provincia	5 vars
CNMC	CNMC (Comisión Nacional de los Mercados y la Competencia)	ANNUAL	Provincia	8 vars
INE	INE (Instituto Nacional de Estadística)	ANNUAL	Provincia	8 vars
MITECO	MITECO (Ministerio para la Transición Ecológica)	ANNUAL	CCAA	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
CCNC	CCN-CERT (Centro Criptológico Nacional)	ANNUAL	National	2 vars
PROT	Protección Civil / DG Catástrofes	ANNUAL	Provincia	2 vars
CSN	CSN (Consejo de Seguridad Nuclear)	ANNUAL	Site-level	1 var
INE-E	INE Economic Statistics	ANNUAL	CCAA	3 vars
REE-S	REE Statistical Series	ANNUAL	CCAA	2 vars
ADIF	ADIF Rail Statistics	ANNUAL	Provincial	1 var
DGT	DGT (Dirección General de Tráfico)	ANNUAL	Provincial	1 var
CNMC-M	CNMC Monitoring Report	ANNUAL	DSO-level	2 vars
IGN	IGN/IGME (Instituto Geográfico Nacional)	STATIC	Provincia	3 vars
CDS	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars

GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
NCSE	NCSE-02 Seismic Code	STATIC	Municipal	1 var
COG-ES	INE Municipal Registry	STATIC	Provincial	1 var
IS09223	ISO 9223 Corrosion	DERIVED	Provincia	1 var
REE	REE (Red Eléctrica de España) — Transparency	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
AEMET	AEMET (Agencia Estatal de Meteorología)	HOURLY	Station	3 vars

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot

FLEET SIZE

3,529

substations scored

MEDIAN R

0.613

P5=0.254 · P95=0.950

HIGH + CRITICAL

2544

72.1% of fleet

HIGH CONFIDENCE

72%

2526 subs · CI/R < 0.50

BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **3,529 substation scores remain unchanged** this edition. The fleet median R of 0.613 (mean 0.614) reflects a distribution skewed toward the lower bands, with 4.3% in Low and 23.6% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

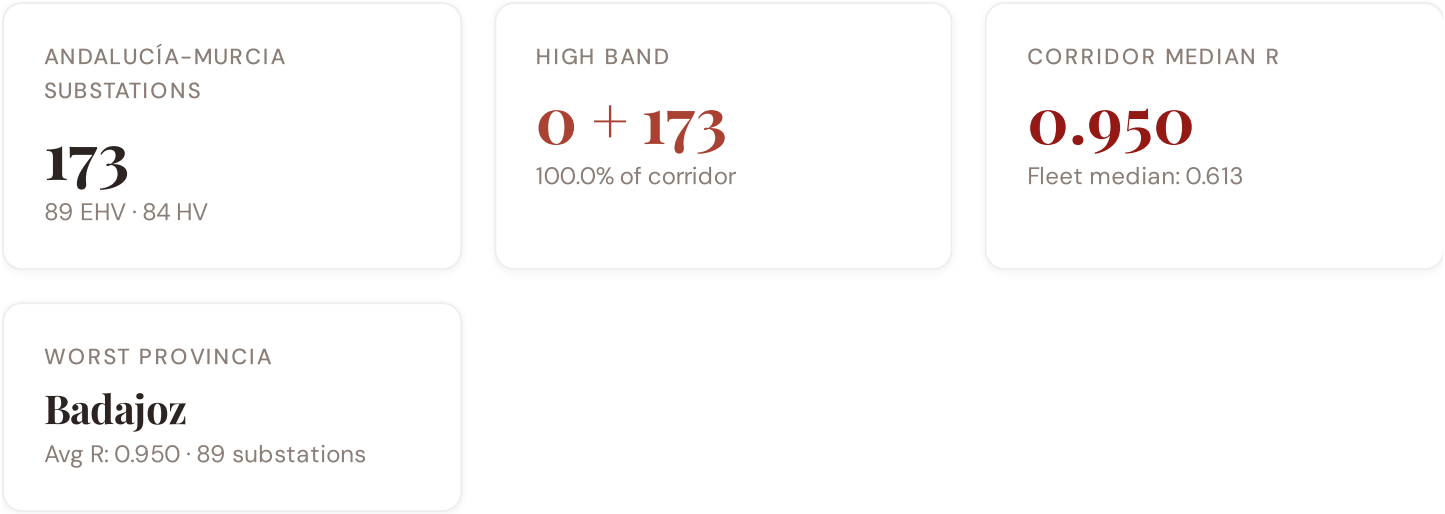
17 changes tracked across PO, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	\$2, \$4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	\$5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	\$5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	\$5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	\$5
F6	ENHANCED	R7 Digital Readiness — Provincia-level DESI model	\$5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	\$2, \$8
L2	ENHANCED	E2 Innovation Enrichment — R&D + economic structure	\$6
L3	ENHANCED	V_socio Fiscal Enrichment — INE + MITECO + energy price	\$5
L4	ENHANCED	R3 Demographic Shift Amplifier — INE net migration	\$5
L5	DATA	95/95 variables operational (100%). 30 data sources total.	\$8, \$12
G1	DATA	IDAE upgraded to quarterly registry — Provincia-level DER registry	\$12
G2	DATA	IGN upgraded to live API — Provincia-level seismic data	\$12
G3	DATA	OSM upgraded to Overpass API — 3,529 real substations	\$12
G4	NEW	R6b Seismic Hazard modifier — NCSE-O2 classification from IGN	\$5
G5	NEW	Network & Topology layer (H): 7 variables — REE Grid Plan, seismic analysis	\$12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	\$12

The Andalucía–Murcia Seismic Corridor: Where Seismic Risk Meets Energy Transition Pressure

Deep–dive rotation: Each edition features a substantive thematic analysis. The 12–month rotation: **Ed. 001** Andalucía–Murcia seismic corridor · **002** DER saturation hotspots (Valencia/Murcia) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Nuclear fleet dependency and phase–out risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T–component forward projections · **009** Provincia Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Andalucía–Murcia, Why Now



The Extremadura hosts **173 substations** across 2 region regions, making it the locus of this edition's deep–dive analysis. Its risk profile is concentrated in the upper bands: **0 substations (0.0%)** are classified High and **173** Critical, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.613. The corridor median R of 0.950 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity–of–supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
STR Santa Amália	Badajoz	0.950	Critical	0.683	0.852	0.581	0.443	0.809	0.824	C, V, S, T
Subestación de Talarrubias	Badajoz	0.950	Critical	0.679	0.940	0.578	0.447	0.789	0.466	C, V, S
STR La Roca de la Sierra	Badajoz	0.950	Critical	0.683	0.919	0.596	0.521	0.784	0.814	C, V, S, T
Subestación OSM 144778695	Badajoz	0.950	Critical	0.659	0.889	0.584	0.562	0.785	0.906	C, V, S, T
Subestación OSM 1486530145	Badajoz	0.950	Critical	0.666	0.957	0.589	0.458	0.814	0.770	C, V, S, T
SET Los Juncales	Badajoz	0.950	Critical	0.670	0.970	0.563	0.404	0.791	0.884	C, V, S, T
Subestación Santa Marina	Badajoz	0.950	Critical	0.700	0.983	0.610	0.491	0.763	0.854	C, V, S, T
ST Alburquerque	Badajoz	0.950	Critical	0.706	0.904	0.572	0.481	0.810	0.742	C, V, S, T
STR Medellín	Badajoz	0.950	Critical	0.676	0.834	0.590	0.468	0.817	0.650	C, V, S

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Canal del Zújar I	Badajoz	0.950	Critical	0.666	0.889	0.542	0.422	0.819	0.774	C, V, S, T
... 163 additional substations — explore full corridor in Digital Twin →										

D.3 Component Analysis

COMPONENT AVERAGES — EXTREMADURA VS FLEET

C — Continuity	0.681 vs 0.481 (+42%)
I — Infrastructure	0.574 vs 0.517 (+11%)
S — Saturation	0.779 vs 0.319 (+145%)
V — Voltage	0.910 vs 0.512 (+78%)
E — Economic	0.475 vs 0.339 (+40%)
T — Transition	0.796 vs 0.663 (+20%)

MODIFIER AVERAGES — EXTREMADURA VS FLEET

R3 Consequence	1.282	fleet 1.119	↑
R4 Graph Crit.	1.011	fleet 1.008	↑
R6a Restoration	1.022	fleet 0.995	↑
R6b Seismic	1.249	fleet 1.200	↑
R7 Digital Read.	0.991	fleet 1.008	→

The dominant risk driver across the Extremadura corridor is **T (Transition)**, averaging 0.796 against a fleet average of 0.663 — occupying 1593% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.910 vs fleet 0.512 (910% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.282 (fleet: 1.119), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.249, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Provincia Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Badajoz	89 subs · avg R = 0.950 · H:0 C:89 M:0
Cáceres	84 subs · avg R = 0.950 · H:0 C:84 M:0

The region regions-level breakdown reveals significant intra-regional variation. **Badajoz** leads with a mean R of 0.950 across 89 substations, while **Cáceres** scores 0.950 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Extremadura is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

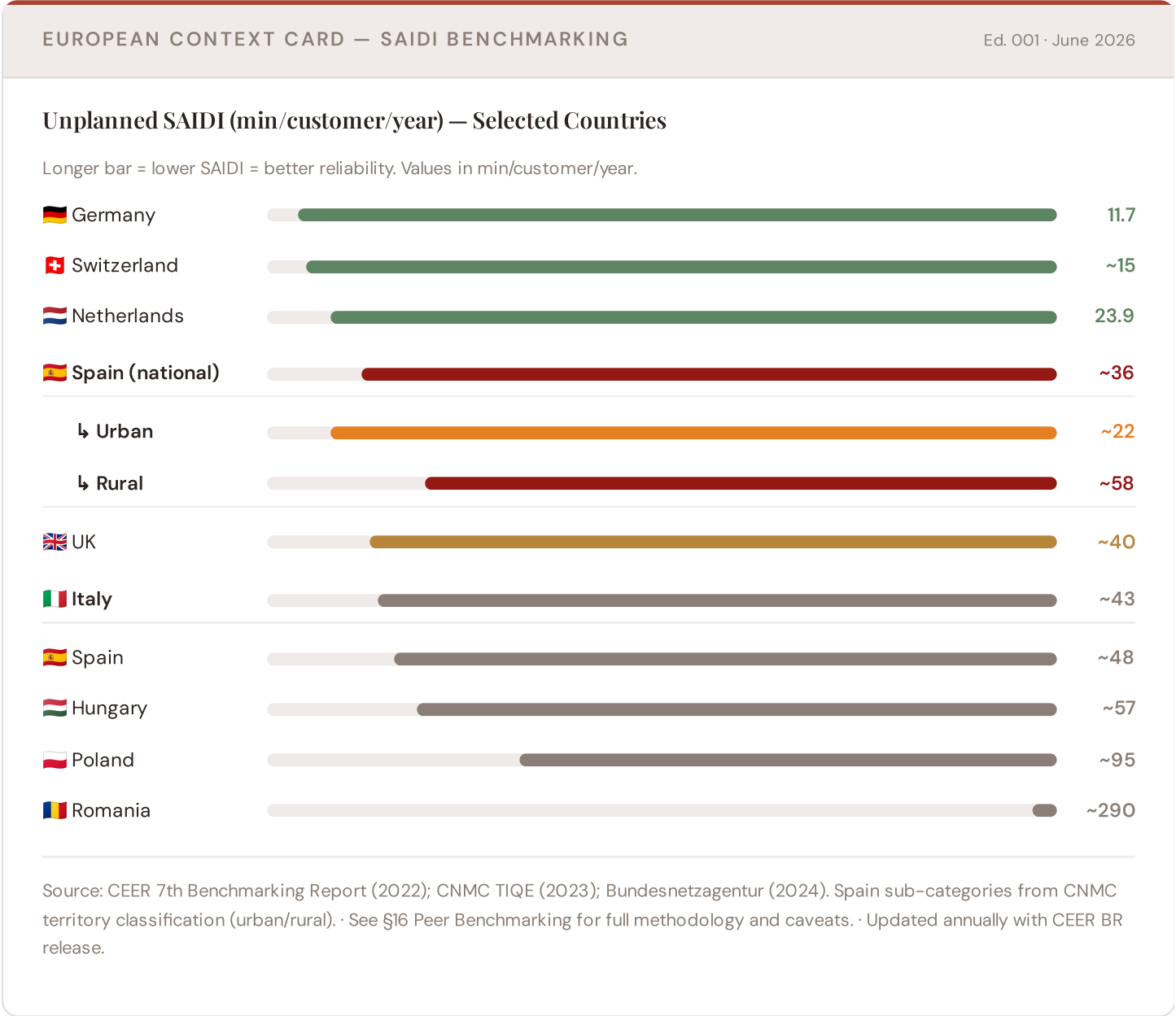
173 Extremadura substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Badajoz sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve

are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Andalucía-Murcia corridor analysis hinges on Spain's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.



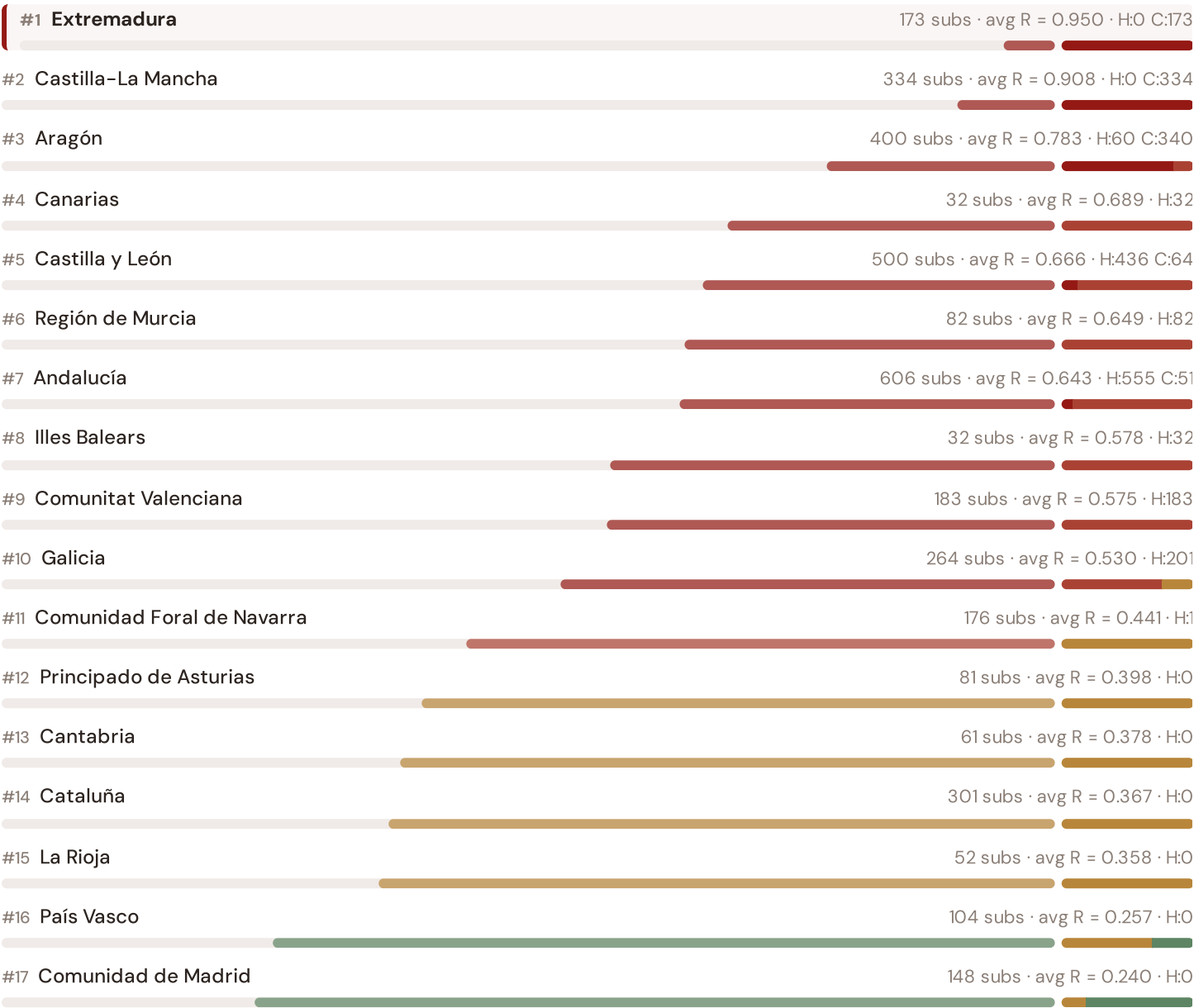
This month's key insight: The Extremadura corridor deep-dive shows **173 substations** (of 173) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory

classification. The corridor's average C of 0.681 is **1.4× the fleet average** of 0.481. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 17 Comunidades Autónomas

Where does the Andalucía-Murcia sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

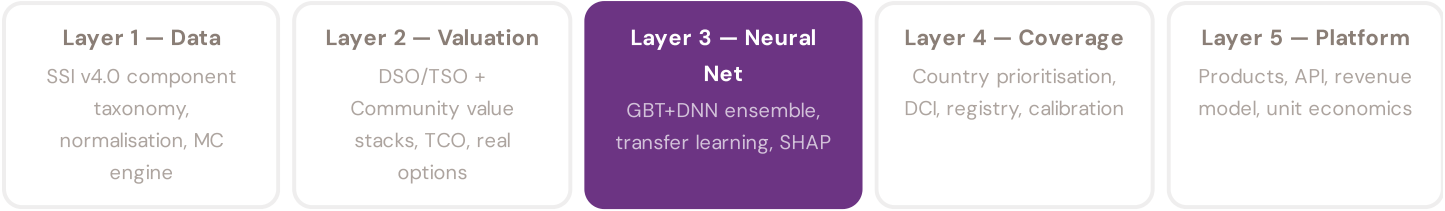
Longer bar = lower R = better resilience. Sorted worst → best.



The Extremadura ranks **#1 of 17 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Extremadura, Castilla-La Mancha, Aragón, while the three most resilient are Comunidad de Madrid, País Vasco, La Rioja. 7 of 17 score above the fleet average of 0.614. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the REE/DSO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 3,529 substations with complete feature vectors. Average CI_width of 0.265 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 153 Low, 832 Medium, 1582 High, 962 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3

Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 1 Quarterly Sources

DGT are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **CNMC (Comisión Nacional de los Mercados y la Competencia)** (8 variables → C1–C4 (TIEPI/NIEPI/CAIDI), quality regulation).

FLEET SPOTLIGHT

A Coruña: Highest Risk Concentration

A Coruña has the highest concentration of High/Critical substations: **103 of 103 (100%)** are in the upper risk bands — avg R = 0.557. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Spanish utility, CNMC, REE, regional DSO, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 003

Published 11 June 2026 · SSI Index v4.0.2 · 3,529 substations · ~22,765 grid lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5